

Dr. Jon R. Fox

Applied Physicist with 25 years of post-doctoral experience spanning MEMS device design and fabrication, model-based systems engineering, and signal analysis and processing, including multiphysics modeling, hardware test, and digital engineering environments. Programmer and data wrangler for microcontrollers, embedded systems, automated test hardware, workstations, and cloud-based servers.

Summary of Skills

- Ph.D. in experimental condensed-matter physics.
- Division Chief Scientist, Architecture and System Development Division of the Applied Systems Laboratory (GTRI).
- Model-Based Systems Engineering with OMG SysML (v1 and v2), MagicDraw/Cameo Enterprise Architecture, and FACE conformance tooling.
- Open-source developer: author of SysML v2 tooling in Python (sysmlpy, uml2py, sysml_canvas) with international collaborators.
- 25+ years of data acquisition and analysis using Python, C/C++, FORTRAN (77/90/95), Go, Groovy, Java, LabVIEW.
- Adept user (and admin) of Unix (Linux, Solaris) and Windows environments, office software suites, and SQL and Graph databases.

Experience and Accomplishments

January 2024 - Present

Division Chief Scientist, Architecture and System Development Division, Applied Systems Laboratory, Georgia Tech Research Institute, Huntsville, AL.

- Technical direction, review, and mentoring across division programs in model-based systems engineering, architecture, and system development.

September 2019 - Present

Principal Research Scientist, Architecture and System Development Division, Applied Systems Laboratory, Georgia Tech Research Institute, Huntsville, AL.

- Model Based Systems Engineering technical analyses for US Army Aviation Future Vertical Lift systems and software architecture.
- GTRI Technical Lead for Future Long-Range Assault Aircraft (FLRAA) architectural risk reduction, utilizing the Future Vertical Lift Architectural Framework (FAF).
- Open-source SysML v2 tooling: clean-room Python implementations of the UML 2.5.1 metamodel (uml2py), a SysML v2 reference API and conformance toolchain (sysmlpy, 123/123 OMG XPECT conformance), an interactive diagramming canvas (sysml_canvas), and a SysML v2 model of the AMS Government Reference Architecture.
- Clean-room Python ports of the FACE Conformance Test Suite data tooling, validated byte-for-byte against the reference Java tools.
- Jython and Groovy scripting for data extraction and addition into Cameo Enterprise Architecture; FACE data modeling with the FAME Awesum tool; member of the Architecture Collaboration Working Group (ACWG).

March 2010 - August 2019

System and Software Architect, Project Manager, Applied Scientist, General Atomics Electromagnetic Systems Group, Huntsville, AL.

- SysML modeling of arresting gear, electromagnetic railgun, and laser effect systems using MagicDraw, integrated with Windchill product lifecycle management.
- Modeling and simulation of linear displacement transducer systems for requirements verification and design downselect.
- Analysis and modeling of pressure data for real-time health prognostication.
- Automated generation of software design documentation from SysML models using Velocity Templating Language (VTL) and Ruby scripting.
- Optimization of hypersonic flight trajectories with NASA's OTIS tool.
- Information and network security analysis with CompTIA Security+ certification.
- MEMS and nanomaterials development at the Charles Bowden Weapons Sciences Laboratory (AMRDEC-WDI): nano-composite piezoelectric microphones, micro-machined aluminum micro-mirrors, and MEMS health transducers; SETA program and functional management until Mar. 2018.

July 2007 - March 2010

Thin Film Process Engineer, Gulton, Inc., South Plainfield, NJ.

- Developed thin film thermal print head (TPH) product lines from prototype to production; process control of existing thick film TPHs.
- Selection, installation, and maintenance of production physical vapor deposition equipment; dry and wet etch process development.
- Automated test equipment for resistor testing and trimming at the die and packaged part level; web application servers for SPC data collection and analysis across the factory floor.

April 2000 - July 2007

Principal Scientist, Research Support Instruments, Princeton, NJ.

- Principal Investigator on SBIR/STTR grants from AFRL, ONR, NIST, and NASA for micro-fabricated structures: micro-machined mass spectrometer, energy-mass spectrograph for micro-satellites, micro-electroformed cross capacitor, and passive MEMS tagging.
- Visiting Professional Technical Staff Member II, Princeton University Department of Mechanical and Aerospace Engineering (2000–2006); Industrial Partner, Princeton Institute for the Science and Technology of Materials.

1993 - 2000

Research Assistant, Penn State Physics Department, University Park, PA.

- Analytical techniques for materials and surfaces: Raman spectroscopy, photoluminescence, HREELS, UPS/XPS/AES, LEED/RHEED, and powder neutron diffraction across fullerenes, Si/Ge/C nanoparticles, ferromagnetic oxides, and II-VI epitaxial films.
- High and ultra-high vacuum deposition: rf & dc magnetron sputtering, e-beam evaporation, and pulsed laser deposition.

Education

Ph. D. Physics

The Pennsylvania State University, University Park, PA, May 2000. Thesis: “*in situ* studies of the electronic and vibrational properties of thin films and novel materials”.

B.S. Physics

Magna cum laude. University of Missouri-Rolla, Rolla, MO, 1993. Minor in mathematics.

Memberships and Certifications

- IEEE Computer Society; the International Council on Systems Engineering (INCOSE).
- INCOSE Certified Systems Engineering Professional (CSEP), March 2025.